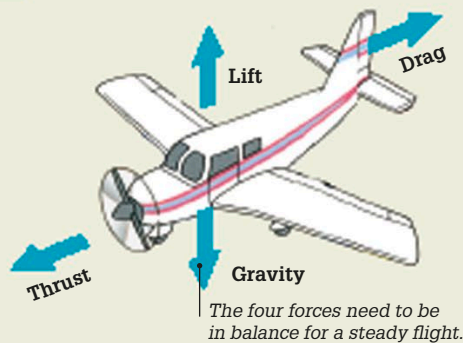


Plane fever

Wilbur Wright toured Europe in 1908–1909, making more than 200 flights and inspiring aviation fever. Dozens of plane makers sprang up, and innovations led to the first seaplane in 1910 (built by French aviator Henri Fabre) and the first four-engined aircraft in 1913 (the Le Grand, built by Russian-American Igor Sikorsky). World War I (1914–1918) saw aircraft manufactured in large numbers, mostly out of a wooden frame covered in stretched linen or canvas.

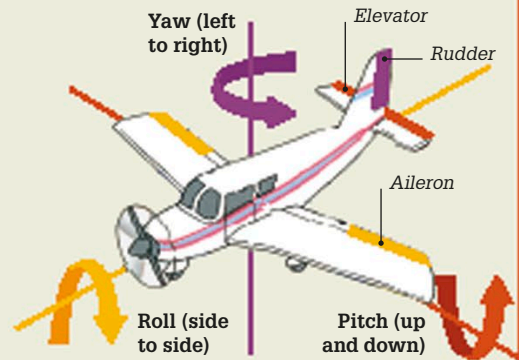


How planes fly



Four forces of flight

As the plane starts moving, air flowing over its wings creates the force of lift. This must exceed the force of gravity for the plane to take-off. Once in the air, the force of thrust from the engine must exceed the force of drag (wind resistance) for the aircraft to move forward.



Yaw, pitch, and roll

Once in the air, an aircraft can move in one of three axes using hinged panels called control surfaces. Ailerons on the wings roll the plane from side to side. On the tail, a rudder yaws (turns) the plane from left to right and elevators pitch it up and down.



Golden age

Aviation blossomed in the era after World War I with a range of peacetime applications, from the spraying of farm fields by cropduster planes to aerial surveying and airmail services. Engines became more powerful and reliable, enabling larger aircraft to be built to carry cargo and passengers. One passenger plane was the Douglas Sleeper Transport (above), launched in 1935, which carried 14 overnight passengers.

Flying science

RADAR

Short for Radio Detection And Ranging, RADAR transmits radio waves, which bounce back off solid objects in order to detect aircraft and missiles.



World War II RADAR installation, Germany

Ejection seats

An explosive charge or a rocket motor in the seat launches a pilot out of the stricken aircraft (usually military) in an emergency and allows him or her to parachute to safety.

Autopilot

The first autopilot was devised by American aviator Lawrence Sperry in 1912. This device keeps a plane flying on a course by adjusting control surfaces and engine power automatically.

1938

The first airliner with a pressurized cabin, the Boeing 307 Stratoliner, flew 33 passengers. It reached an altitude of 20,000 ft (6,000 m), well above weather disturbances to cruise more smoothly and at higher speeds.

1939

The maiden voyage of the Heinkel He-178 became the first flight of an aircraft powered by a jet engine. It was built by German aircraft designer Ernst Heinkel.



1952

The de Havilland Comet became the first jet airliner to enter service with an airline, BOAC (British Overseas Airways Corporation). Its four jet engines gave it a range of up to 1,490 miles (2,400 km).

1957

The first business jet—the Lockheed JetStar—took flight. It seated ten passengers and two crew. Some 204 JetStars would be built.

